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LAB10 – INTRODUCTION TO BISON

Sample

Sample.l

```
%{
   #include "sample.tab.h"
%}

%%
[0-9]+ {return NUMBER;}
\n {return NL;}
[a-zA-Z][a-zA-Z0-9_]* {return ID;}
. {return yytext[0];}
%%

int yywrap(){
   return 1;
}
```

Sample.y

```
%{
   #include<stdio.h>
   #include<stdlib.h>
```

```
   int yylex(void);
   int yyerror(char *msg);
%}

%token NUMBER ID NL
%left '+'
%left '*'

%%
stmt : exp NL {printf("Valid Expression");exit(0);};

exp : exp '+' term | term;
term : term '*' factor | factor;
factor : ID | NUMBER;
%%

int yyerror(char *msg){
   printf("Invalid Expression\n");
   exit(0);
}

int main(){
   printf("Enter the Expression\n");
   yyparse();
   return 0;
}
```

OUTPUT -

```
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ bison -d sample.y
sample.y:19 parser name defined to default :"parse"
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ flex sample.l
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ gcc lex.yy.c sample.tab.c -o sample
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./sample
Enter the Expression
3*25+100
Valid Expression
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./sample
Enter the Expression
3*+5
Invalid Expression
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$
```

Q1. To check a valid declaration statement.

Q1.l

```
%{
  #include "q1.tab.h"
%}

%%
int|float|char|double {return TYPE;}
[a-zA-Z][a-zA-Z0-9_]* {return ID;}
"," {return COMMA;}
";" {return SEMICOLON;}
[ \t\n] ;
. {return yytext[0];}
%%

int yywrap(){
  return 1;
}
```

Q1.y

```
%{
   #include<stdio.h>
   #include<stdlib.h>

   int yylex(void);
   int yyerror(char *msg);
%}

%token TYPE ID COMMA SEMICOLON NL

%%
stmt : decl {printf("Valid Declaration\n");exit(0);};

decl : TYPE list SEMICOLON;

list : list COMMA ID
    | ID;
%%

int yyerror(char *msg){
   printf("Invalid Declaration\n");
   exit(0);
}

int main(){
   printf("Enter the Declaration\n");
   yyparse();
   return 0;
}
```

OUTPUT -

```
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ bison -d q1.y
q1.y:18 parser name defined to default :"parse"
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ flex q1.l
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ gcc lex.yy.c q1.tab.c -o q1
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./q1
Enter the Declaration
int a,b,c;
Valid Declaration
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./q1
Enter the Declaration
int a, float b;
Invalid Declaration
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$
```

Q2. To check a valid decision making statements.

Q2.l

```
%{ #include "q2.tab.h" %}
%% if {return IF;} else {return ELSE;} [0-9]+ {return NUMBER;} [a-zA-Z][a-zA-Z0-9_]* {return ID;} "=="|"<="|">="|"!="|"<"|">" {return RELOP;}
[ \t] ; /* ignore spaces and tabs */
\n {return NL;}
. {return yytext[0];} %%
int yywrap(){ return 1; }
```

Q2.y

```
%{
  #include<stdio.h>
  #include<stdlib.h>

  int yylex(void);
  int yyerror(char *msg);
%}

%token IF ELSE ID NUMBER RELOP NL
```

```
%%
stmt : decision NL {printf("Valid Decision Statement\n");exit(0);};

decision : IF '(' cond ')' statement
       | IF '(' cond ')' statement ELSE statement;

statement : ID ';'
        | ID '=' NUMBER ';';

cond : ID RELOP ID
    | ID RELOP NUMBER;
%%

int yyerror(char *msg){
   printf("Invalid Decision Statement\n");
   exit(0);
}

int main(){
   printf("Enter the Decision Statement\n");
   yyparse();
   return 0;
}
```

OUTPUT -

```
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ bison -d q2.y
q2.y:24 parser name defined to default :"parse"
q2.y contains 1 shift/reduce conflict.
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ flex q2.l
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ gcc lex.yy.c q2.tab.c -o q2
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./q2
Enter the Decision Statement
if(a>b) x; else if(a<b) y; else z;
Valid Decision Statement
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./q2
Enter the Decision Statement
if(a<b) z; else x; else y;
Invalid Decision Statement
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$
```

Q3. To evaluate an arithmetic expression involving operations +,-,* and /.

Q3.l

```
%{
  #include "q3.tab.h"
%}

%%
[0-9]+ {yylval=atoi(yytext); return NUMBER;}
\n {return NL;}
[+\-*/] {return yytext[0];}
. {return yytext[0];}
%%

int yywrap(){
  return 1;
}
```

Q3.y

```
%{ #include<stdio.h> #include<stdlib.h>
```

```
int yylex(void);
int yyerror(char *msg);

%}
%token NUMBER NL %left '+''-' %left '*''/'
%% stmt : exp NL {printf("Result = %d\n",$1);exit(0);};
exp : exp '+' exp {$$=$1+$3;} | exp '-' exp {$$=$1-$3;} | exp '' exp {$$=$1$3;} |
exp '/' exp {$$=$1/$3;} | NUMBER {$$=$1;} ; %%
int yyerror(char *msg){ printf("Invalid Expression\n"); exit(0); }
int main(){ printf("Enter the Expression\n"); yyparse(); return 0; }
```

OUTPUT -

```
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ bison -d q3.y
q3.y:22 parser name defined to default :"parse"
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ flex q3.l
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ gcc lex.yy.c q3.tab.c -o q3
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./q3
Enter the Expression
5+2-3
Result = 4
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./q3
Enter the Expression
10*10/2
Result = 50
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./q3
Enter the Expression
3*+2
Invalid Expression
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$
```

Q4. To validate a simple calculator using postfix notation. The grammar rules are as follows – input → input line | ε

line → '\n' | exp '\n'

exp → num | exp exp '+'

| exp exp '-'

```
| exp exp '*'
| exp exp '/'
| exp exp '^'
| exp 'n'
```

Q4.l

```
%{
   #include "q4.tab.h"
%}

%%

[0-9]+ {return NUM;}

[ \t] ;

\n {return NL;}

[+\-*/^n] {return yytext[0];}

. {return yytext[0];}

%%

int yywrap(){
   return 1;
}
```

Q4.y

```
%{
   #include<stdio.h>
   #include<stdlib.h>
```

```
   int yylex(void);
   int yyerror(char *msg);
%}

%token NUM NL

%%

input : line {exit(0);};

line : exp NL {printf("Valid Postfix Expression\n");};

exp : NUM
   | exp exp '+'
   | exp exp '-'
   | exp exp '*'
   | exp exp '/'
   | exp exp '^'
   | exp 'n'
   ;

%%

int yyerror(char *msg){
   printf("Invalid Postfix Expression\n");
   exit(0);
}

int main(){
   printf("Enter Postfix Expression\n");
   yyparse();
   return 0;
}
```

OUTPUT -

```
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ bison -d q4.y
q4.y:26 parser name defined to default :"parse"
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ flex q4.l
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ gcc lex.yy.c q4.tab.c -o q4
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./q4
Enter Postfix Expression
2 3 +
Valid Postfix Expression
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$ ./q4
Enter Postfix Expression
2 + 3 -
Invalid Postfix Expression
6CSEC1@dbl-29:~/Desktop/Yash Chauhan/Lab10$
```